A260 – The Casimir Effect in FFGFT: Vacuum Structure and Experimental Tests

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July 2026

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.1 Purpose

[STIPULATION] This document treats the Casimir effect within the FFGFT framework. The Casimir effect is in the legacy corpus the only experimentally accessible bridge between the fundamental length scale $L_0 = \xi \cdot \ell_P$ and measurable vacuum forces. The document shows: (1) The modified Casimir formula reproduces the standard formula exactly – no new physics at currently accessible separations. (2) The prefactor $4/3$ in $\xi = 4/3 \times 10^{-4}$ is confirmed by the Casimir effect independently of mass fits (A015). (3) There is a characteristic vacuum length scale $L_\xi \approx 100\,\mu$m at which Casimir energy density and a specific vacuum quantity coincide – this is here explicitly excluded.

.2 Fundamental Length Scales

[B] FFGFT defines a hierarchy of characteristic length scales:

$$L_0 = \xi \cdot \ell_P \approx 2.155 \times 10^{-39}\,\text{m} \quad \text{(sub-Planck, minimal spacetime granulation)},$$

$$\ell_P = \sqrt{\hbar G/c^3} \approx 1.616 \times 10^{-35}\,\text{m} \quad \text{(Planck length)},$$

$$L_\xi \approx 100\,\mu\text{m} \quad \text{(characteristic vacuum length scale, measurable)}.$$

The coupling constant $\xi = 4/3 \times 10^{-4}$ is the same parameter as in A020. It can be derived from a Lagrangian describing the dynamics of the time field (A142).

[B] Spacetime granulation. At L_0 all vacuum fluctuations are fully active. For $d > L_0$ only parts of these forces are visible – this is the physical origin of the $1/d^4$ dependence of the Casimir force. Due to the extremely small size of $L_0 \approx 2 \times 10^{-39}$ m, direct measurement is not possible.

.3 The Modified Casimir Formula and Its Consistency

[K] The modified Casimir energy density in the FFGFT framework:

$$|\rho_{\text{Casimir}}(d)| = \frac{\pi^2}{240\,\xi}\,\rho_{\text{vac}} \cdot \left(\frac{L_\xi}{d}\right)^4,$$

where $\rho_{\text{vac}} = \xi\hbar c/L_\xi^4$ is a characteristic vacuum density. Substituting:

$$|\rho_{\text{Casimir}}(d)| = \frac{\pi^2}{240\,\xi} \cdot \frac{\xi\hbar c}{L_\xi^4} \cdot \frac{L_\xi^4}{d^4} = \frac{\pi^2\hbar c}{240\,d^4}.$$

This is exactly the standard Casimir formula. No contradiction, no new parameter – the FFGFT formulation is numerically equivalent to the established QED results.

[K] Numerical verification (Casimir energy density vs. plate separation):

Separation d	ρ_{Casimir} (J/m³)	Ratio to ρ_{vac}
$100\,\mu$m	4.17×10^{-14}	1.00
$10\,\mu$m	4.17×10^{-10}	10^4
$1\,\mu$m	4.17×10^{-2}	10^{12}

[K] Characteristic scale hierarchy:

$$\frac{L_0}{\ell_P} = \xi = 1.333 \times 10^{-4}, \qquad \frac{\ell_P}{L_\xi} \approx 1.6 \times 10^{-31}.$$

.4 Confirmation of the Factor $4/3$

[B] The prefactor $4/3$ in $\xi = 4/3 \times 10^{-4}$ appears independently in the Casimir context: the factor $4/3$ appears in the Casimir energy term as $E_{\text{Casimir}} \propto \frac{4}{3}$ (the fourth – musical interval $4:3$) and is therefore not only a mass-fit quantity. This independence is recorded in A015 as Casimir confirmation: no other prefactor (fifth $3/2$, third $5/4$) gives consistent results for the mass spectrum *and* the Casimir structure simultaneously.

.5 Experimental Tests

[K] Three classes of tests:

Precision measurements at $d \approx L_\xi$: At plate separation $\approx 100\,\mu$m the Casimir energy density reaches the order of ρ_{vac}. This regime is experimentally accessible.

Scaling behaviour: The $1/d^4$ dependence should be exactly satisfied down into the nanometre range. Deviations at $d \lesssim 10$ nm could indicate spacetime granulation.

Experimental L_ξ values from the literature: Parallel plates: ~ 228 nm; sphere-plate geometry: $\sim 1.75\,\mu$m to $\sim 18\,\mu$m. The scatter reflects geometric differences ($F \propto 1/d^4$ vs. $F \propto 1/d^3$).

.6 Verification Script

- Numerical verification of the modified Casimir formula (reproduction of the standard formula), length scale hierarchy, comparison with L_ξ values: `python/A_Serie_Skripte/a260_casimir.py`

.7 Open Questions

Direct measurement of L_0 is not possible. The minimal length scale $L_0 \approx 2 \times 10^{-39}$ m lies far beyond any experimental reach. The characteristic scale L_ξ is experimentally accessible, but its precise connection to L_0 presupposes the FFGFT framework.

Deviations from the $1/d^4$ law at $d \sim 10$ nm are not measured. The prediction of a scale change near the sub-Planck boundary is qualitative, not quantitatively worked out.

.8 Supersedes

This document subsumes: Dok. 091 (modified Casimir theory, length scale hierarchy, numerical verification, experimental predictions and tests). *Not* taken over: the Casimir vacuum scaling from Dok. 091 (excluded sector, outside A-series core) and the macroscopic vacuum implications.

.9 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests with the author. Full wording of the rules in A010.